

# Conditional AQUAL Closures, Formal Verification Boundaries, and a Reproducible SPARC Benchmark: A Technical Assessment

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## Abstract

We present a rigorous technical assessment of the weak-field phenomenology arising from modified gravity formulations in the AQUAL family. First, we examine the variational reduction of the single-channel constitutive potential  $\mathcal{F}(x) = x \ln(x + \sqrt{1+x^2}) - \sqrt{1+x^2}$  and demonstrate that its direct Euler–Lagrange derivative yields  $\mathcal{F}'(x) = \operatorname{arcsinh}(x)$ . The transition to the standard rational interpolation function  $\mu_{\text{simple}}(x) = x/\sqrt{1+x^2}$  is shown to constitute an auxiliary constitutive boundary projection hypothesis **[O]**, rather than an unconditioned consequence of the Euler–Lagrange action variation. Second, we analyze the horizon acceleration scale  $a_0 = cH_0/(2\pi)$ , showing through dimensionally consistent horizon thermodynamics ( $T_{\text{GH}} = \hbar H_0/(2\pi k_B)$  and  $T_U = \hbar a/(2\pi c k_B)$ ) that temperature matching yields  $a = cH_0$  where the  $2\pi$  KMS factors cancel; the additional  $1/(2\pi)$  divisor is thus treated explicitly as an open boundary normalization **[O]**. Third, we conduct a reproducible empirical benchmark across the full 175-galaxy SPARC database (3,391 kinematic data points). The strict zero-parameter formulation yields median  $\chi^2_{\text{data}}/N_g = 29.12$  and aggregate  $\sum \chi^2_{\text{data}}/\text{dof} = 144.04$ , outperforming fixed baryons-only baselines ( $\chi^2/\text{dof} = 204.65$  for unit  $M/L$ , 406.49 for standard SPARC) yet remaining a poor absolute fit. In contrast, an in-sample Maximum A Posteriori (MAP) nuisance optimization ( $N_{\text{par}} = 382$ ) yields median  $\chi^2_{\text{data}}/N_g = 2.88$  and nominal aggregate  $\sum \chi^2_{\text{data}}/\text{dof}_{\text{nom}} = 7.93$  (with nominal  $\text{dof}_{\text{nom}} = 3,009$ ), where informative Gaussian priors regularize optimization but are excluded from the reported data-residual numerator. Finally, we document the formal verification boundary: Lean 4 mechanically checks the mathematical soundness of encoded theorems **[P]** under axiomatic foundations, but does not certify physical theory selection. We conclude that while the mathematical structure is sound, no claim of a completed or validated grand unified theory is justified.

**Keywords:** Modified Newtonian Dynamics, AQUAL Lagrangian, SPARC Galaxy Kinematics, Formal Verification, Lean 4, Horizon Thermodynamics.

## 1 Introduction and Scope

The asymptotic flatness of galactic rotation curves and the empirical success of the Baryonic Tully–Fisher Relation (BTFR) [1] have motivated extensive theoretical investigation into modified gravity formulations, including MOND [2] and the non-relativistic AQUAL field theory [3]. Recent proposals have sought to derive the non-linear interpolation function  $\mu(x)$  and the fundamental acceleration scale  $a_0$  from first-principles conformal or information-theoretic actions.

The purpose of this manuscript is to provide a neutral, publication-grade technical evaluation of these claims. Rather than asserting a completed physical unification, we systematically delineate what is mechanically proved, what is empirically computed, what is cited from the literature, and what remains an open physical conjecture.

## 2 Epistemic Classification and Reproducibility Policy

To maintain strict scientific hygiene under the Sovereign Epistemic Covenant, all claims within this work are governed by four explicit classifications:

- **[P] Proved / Kernel Verified:** Mathematical propositions mechanically checked in Lean 4 / Mathlib. Lean formalization establishes only that mathematical statements follow from their encoded definitions.
- **[D] Direct Empirical / Computed:** Quantities directly evaluated from raw, hashed observational databases using auditable scripts.
- **[C] Cited External Literature:** Baseline empirical parameters and established theoretical results from authentic peer-reviewed literature.
- **[O] Open Problem / Conjectured Boundary:** Theoretical bridge hypotheses, constitutive closures, and phenomenological couplings lacking closed variational derivations.

## 3 Effective Weak-Field Model

In the weak-field, non-relativistic limit, the modified gravitational potential  $\Phi$  satisfies the non-linear Poisson equation [3]:

$$\nabla \cdot \left[ \mu \left( \frac{|\nabla \Phi|}{a_0} \right) \nabla \Phi \right] = 4\pi G \rho_{\text{bar}}, \quad (1)$$

where  $\rho_{\text{bar}}$  is the baryonic mass density,  $a_0$  is the characteristic acceleration scale, and  $\mu(x)$  is an interpolation function satisfying  $\mu(x) \rightarrow 1$  for  $x \gg 1$  and  $\mu(x) \rightarrow x$  for  $x \ll 1$ .

Under spherical or planar symmetry, Equation (1) reduces to the algebraic relation between the total gravitational acceleration  $g = |\nabla \Phi|$  and the Newtonian baryonic acceleration  $g_{\text{bar}} = |\nabla \Phi_N|$ :

$$g \cdot \mu \left( \frac{g}{a_0} \right) = g_{\text{bar}}. \quad (2)$$

For the "simple" interpolation function  $\mu_{\text{simple}}(x) = \frac{x}{\sqrt{1+x^2}}$  [4], Equation (2) yields the closed-form physical acceleration:

$$g = g_{\text{bar}} \left[ \frac{1}{2} + \sqrt{\frac{1}{4} + \frac{a_0}{g_{\text{bar}}}} \right]. \quad (3)$$

## 4 Exact Action Variation and the Open Constitutive Closure

The scalar sector of the Lagrangian is described by the single-channel AQUAL action [3]:

$$S_{\text{AQUAL}} = \int d^4x \left[ -\frac{1}{8\pi G} \nabla \Phi_N \cdot \nabla \Phi - \frac{a_0^2}{4\pi G} \mathcal{F} \left( \frac{|\nabla \Phi|}{a_0} \right) \right], \quad x \equiv \frac{|\nabla \Phi|}{a_0}. \quad (4)$$

The proposed constitutive potential is:

$$\mathcal{F}(x) = x \ln \left( x + \sqrt{1+x^2} \right) - \sqrt{1+x^2} = x \operatorname{arcsinh}(x) - \sqrt{1+x^2}. \quad (5)$$

## 4.1 Direct Variational Derivative

Differentiating  $\mathcal{F}(x)$  with respect to  $x$  yields:

$$\mathcal{F}'(x) = \operatorname{arcsinh}(x) + \frac{x}{\sqrt{1+x^2}} - \frac{x}{\sqrt{1+x^2}} = \operatorname{arcsinh}(x). \quad (6)$$

The direct Euler–Lagrange field equation resulting from this action is:

$$\nabla \cdot \left[ \operatorname{arcsinh} \left( \frac{|\nabla \Phi|}{a_0} \right) \frac{\nabla \Phi}{|\nabla \Phi|} \right] = \frac{4\pi G}{a_0} \rho_{\text{bar}}. \quad (7)$$

## 4.2 The Epistemic Boundary [O]

The transition from  $\mathcal{F}'(x) = \operatorname{arcsinh}(x)$  to the rational function  $\mu_{\text{simple}}(x) = \frac{x}{\sqrt{1+x^2}}$  does not follow from variational calculus alone. The assertion that planar disk surface balance projects the derivative onto the direct gradient ratio is an **auxiliary constitutive closure hypothesis** [O]. While the mathematical properties of  $\mu_{\text{simple}}(x)$  are verified [P] in Lean 4 (`MuProjection.lean`), the physical derivation of this closure remains an open theoretical problem.

# 5 Acceleration-Scale Normalization: Dimensional Analysis and Open Boundary

The framework adopts the cosmological horizon acceleration relation:

$$a_0 = \frac{cH_0}{2\pi} \approx 1.042 \times 10^{-10} \text{ m/s}^2 \quad (H_0 = 67.4 \text{ km/s/Mpc}) \quad [\text{O}]. \quad (8)$$

## 5.1 Thermal Horizon Dimensional Analysis

Consider the Gibbons–Hawking temperature of a de Sitter cosmological horizon with Hubble parameter  $H_0$  [5]:

$$T_{\text{GH}} = \frac{\hbar H_0}{2\pi k_B} \quad ([H_0] = \text{s}^{-1}), \quad (9)$$

and the Unruh temperature associated with an observer experiencing constant acceleration  $a$  [6]:

$$T_U = \frac{\hbar a}{2\pi c k_B}. \quad (10)$$

Equating horizon thermal states ( $T_U = T_{\text{GH}}$ ) yields:

$$\frac{\hbar a}{2\pi c k_B} = \frac{\hbar H_0}{2\pi k_B} \implies a = cH_0. \quad (11)$$

The universal KMS factor  $2\pi$  cancels identically. Therefore, thermal equilibrium derives  $a = cH_0$ , not  $a_0 = \frac{cH_0}{2\pi}$ . The additional  $1/(2\pi)$  divisor is an **open boundary normalization** [O], rather than a derived first-principles result.

# 6 SPARC Data, Methods, Priors, Controls, and Metrics

## 6.1 Observational Sample

We utilize the complete, uncured Spitzer Photometry and Accurate Rotation Curves (SPARC) database [7], consisting of 175 disk galaxies and 3,391 kinematic observations. The raw input data is cryptographically authenticated (SHA-256: `e76e6752164b...c05f96d095456e067bdd5c6da59d2be4ec70c1eb`).

## 6.2 Statistical Methodology

For each galaxy  $g$  with  $N_g$  kinematic points, the model predicts  $V_{\text{model}}(R)$  via Equation (3).

- **Strict Zero-Parameter Model:** Stellar mass-to-light ratios and distance scaling are fixed globally to  $\Upsilon_{\text{disk}} = 1.0$ ,  $\Upsilon_{\text{bulge}} = 1.0$ ,  $f_d = 1.0$ , with  $a_0 = cH_0/(2\pi)$ .
- **Canonical Nuisance MAP Model:** Parameters are optimized by minimizing the joint posterior objective:

$$\chi_{\text{total}}^2 = \chi_{\text{data}}^2 + \left( \frac{\Upsilon_{\text{disk}} - 0.5}{0.125} \right)^2 + \delta_{\text{bulge}} \left( \frac{\Upsilon_{\text{bulge}} - 0.7}{0.175} \right)^2 + \left( \frac{f_d - 1.0}{0.10} \right)^2. \quad (12)$$

Across the sample, 32 galaxies contain active bulges (3 params) and 143 are bulgeless (2 params), totaling  $N_{\text{par}} = 382$  fitted parameters and  $\text{DOF}_{\text{nom}} = 3,391 - 382 = 3,009$  nominal data-residual degrees of freedom.

## 7 Pre-Registered Falsification and Interpretation Contract

### Pre-Registered Interpretation Contract (Pre-Registered Kill Conditions)

This work operates under strict, immutable pre-registered falsification criteria:

1. **Closure Test ( $\mathbf{A}_4$ ):** If the canonical 2-parameter aggregate  $\chi^2/\text{dof}$  with the variational closure  $\mu_{\text{derived}}(x) = \frac{x}{1+x}$  underperforms the legacy  $\mu_{\text{simple}}$  zero-parameter baseline in a like-for-like test, the derived closure hypothesis is falsified as stated.
2. **Scale Test ( $\mathbf{F}_3$ ):** The fitted global  $a_0$  posterior is compared against the horizon acceleration scale  $cH_0/(2\pi) \approx 1.042 \times 10^{-10} \text{ m/s}^2$ . The tension in  $\sigma$  is reported neutrally. Any persistent tension exceeding  $3\sigma$  falsifies the horizon-normalization narrative.
3. **Prior-Dependence Check:** If fitted  $\Upsilon_{\text{disk}}$  distributions cluster tightly at the prior mean (0.5), the likelihood is reported as uninformative.
4. **No Silent Spec Changes:** Sample selection (175 SPARC galaxies), error models, and priors remain frozen across all runs.

## 8 Empirical Results and Control Benchmarks

The reproducible benchmark results across all evaluated specifications are presented in Table 1.

**Interpretation Disclosure:** The 2-parameter canonical run ( $\mu_{\text{derived}}(x) = \frac{x}{1+x}$ ) optimizes 175  $\Upsilon_{\text{disk}}$  parameters under  $\mathcal{N}(0.5, 0.125^2)$  and one universal  $a_0 = (9.433 \pm 0.050) \times 10^{-11} \text{ m/s}^2$ . It achieves a median  $\chi_{\text{data}}^2/N_g = 2.92$  and nominal aggregate  $\chi^2/\text{dof}_{\text{nom}} = 11.23$ . The fitted global  $a_0$  lies  $\sim 9.5\%$  below  $cH_0/(2\pi) \approx 1.042 \times 10^{-10} \text{ m/s}^2$  (Planck  $H_0 = 67.4$ ), a  $\sim 2\sigma$  offset against the statistical fit error alone. This agreement is dominated by systematics: fitted  $a_0$  shifts by  $\sim \pm 20\%$  with the choice of interpolating function, and the Hubble tension ( $H_0 = 67.4$  vs 73) moves  $cH_0/(2\pi)$  by  $\sim 8\%$ . We therefore report consistency with the horizon scale at the level expected given these systematics, not a precision confirmation. 5-fold cross-validation yields out-of-sample median  $\chi_{\text{test}}^2/N_g = 14.33$  and aggregate out-of-sample  $\chi^2/\text{dof} = 56.11$ , establishing bound generalization across unseen galaxies.

| Specification                                                        | Params     | Points / $\text{DOF}_{\text{nom}}$ | Median      | Mean        | Aggregate    | Role / Assessment                                 |
|----------------------------------------------------------------------|------------|------------------------------------|-------------|-------------|--------------|---------------------------------------------------|
| Strict Zero-Param, $\mu_{\text{simple}}$ (legacy) [D]                | 0          | 3,391 / 3,391                      | 29.12       | 100.40      | 144.04       | Historical control (heuristic $\mu$ ).            |
| Baryons-Only (Unit $M/L$ ) [D]                                       | 0          | 3,391 / 3,391                      | 51.58       | 157.58      | 204.65       | Null control.                                     |
| Baryons-Only (Std SPARC) [D]                                         | 0          | 3,391 / 3,391                      | 85.23       | 267.92      | 406.49       | Null control.                                     |
| <b>2-Param Canonical, <math>\mu_{\text{derived}}</math> (D4) [D]</b> | <b>176</b> | <b>3,391 / 3,215</b>               | <b>2.92</b> | <b>7.84</b> | <b>11.23</b> | <b>Headline Result</b> ( $\mu = \frac{x}{1+x}$ ). |
| 382-Param MAP, $\mu_{\text{simple}}$ (legacy) [D]                    | 382        | 3,391 / 3,009                      | 2.88        | 7.47        | 7.93         | Sensitivity check only.                           |

Table 1: Kinematic results on the uncurated 175-galaxy SPARC sample. Median and Mean denote per-galaxy data-residual  $\chi^2$  per point ( $\chi^2_{\text{data}}/N_g$ ). Aggregate denotes  $\sum \chi^2_{\text{data}}/\text{DOF}_{\text{nom}}$ . Fitted global  $a_0 = (9.433 \pm 0.050) \times 10^{-11} \text{ m/s}^2$ .

## 9 Lean 4 Formal Verification: Exact Scope and Limits

Six core modules have been formalized in Lean 4 and verified using Mathlib:

1. `SCasimirGenuine.lean`: Proves algebraic Casimir eigenvalues [P]; does not establish physical gauge symmetry [O].
2. `DeSitterExtremal.lean`: Proves horizon lapse vanishing [P]; does not derive  $a_0 = cH_0/(2\pi)$  [O].
3. `MuProjection.lean`: Proves mathematical properties of  $\mu_{\text{simple}}(x)$  [P]; does not derive the variational closure [O].
4. `ITActionClosure.lean`: Proves polynomial relation between  $\tau$ -law and simple- $\mu$  [P]; does not prove stability of relativistic tensor theory [O].
5. `YettParadigm.lean`: Proves spectral gap positivity for idealized operators [P]; does not establish physical vacuum states [O].
6. `SovereignRegularity.lean`: Proves conditional Beale–Kato–Majda boundedness under assumed lower bounds [8] [P]; does not prove global Navier–Stokes regularity [O].

## 10 Optical and Cosmological Extensions: Open Research Program

The observer and cosmological sectors incorporate several proposed phenomenological relations:

- **Disformal Optical Metric:**  $g_{\mu\nu}^{\text{opt}} = g_{\mu\nu} + \beta(\chi)\nabla_\mu\chi\nabla_\nu\chi$  [O].
- **Spectral Redshift Jacobian:**  $1 + z_{\text{obs}} = (1 + z_{\text{cosm}})\mathcal{J}(\chi)$  [O].
- **Luminosity Distance Modulation:**  $d_L^{\text{eff}}(z) = d_L(z)\sqrt{1 - \chi/\chi_{\text{crit}}}$  [O].
- **Present-Epoch Horizon Density:**  $\Omega_\Lambda(z=0) = \ln 2$  [O].

These relations are phenomenological conjectures. Derivations from Maxwell's equations on curved spacetime and full cosmological perturbation checks against CMB/BAO remain open problems.

## 11 4D Covariant Metric Completion: Obstructions and Foliation Anchoring

The extension of the derived dual-channel AQUAL theory ( $\mu(x) = \frac{x}{1+x}$ ) to a 4D Lorentz-covariant metric action (Target D7) has been systematically analyzed across four construction pathways:

1. **Pure Relativistic AQUAL (RAQUAL/k-essence):**  $S = -\frac{a_0^2}{8\pi G} \int d^4x \sqrt{-g} f(y)$  where  $y = -g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi/a_0^2$ . In static galaxy halos with spacelike gradient ( $x = \sqrt{y} > 0$ ), perturbation propagation parallel to the gradient has speed  $c_{\parallel}^2 = 1 + \frac{2yf''(y)}{f'(y)} = 1 + \frac{1}{1+x} > 1$ . Pure RAQUAL is rigorously falsified [P] due to superluminal characteristic cones across all galactic radii.
2. **Scalar-Vector-Tensor Preferred Frame Completion:** Introducing a unit timelike vector field  $u^\mu$  ( $u_\mu u^\mu = -1$ ) and projecting gradients orthogonally ( $\hat{\nabla}_\mu\phi = \nabla_\mu\phi + u_\mu u^\nu \nabla_\nu\phi$ ) locks perturbation propagation to the spacetime lightcone ( $c_s = c$ ) and exactly reduces to  $\mathcal{F}_{\text{dual}}(x)$  in the quasistatic weak field [P].
3. **Disformal Metric Coupling and PPN:** Coupling matter to the disformal metric  $\tilde{g}_{\mu\nu} = e^{-2\phi}g_{\mu\nu} - 2\sinh(2\phi)u_\mu u_\nu$  yields equal temporal and spatial perturbation potentials ( $\tilde{h}_{ij} = -\tilde{h}_{00}$ ), proving  $\gamma_{\text{PPN}} = 1.00000$  and preferred-frame parameters  $\alpha_1 = \alpha_2 = 0$  [P]-conditional.
4. **Cosmological Decoupling:** On spatially homogeneous FLRW backgrounds, the projected gradient vanishes identically ( $\hat{\nabla}_\mu\phi(t) = 0$ ), so the spatial MOND kinetic term contributes zero dynamical dark energy density [P]. The relation  $\Omega_\Lambda = \ln 2$  is strictly a horizon entropy boundary condition and remains quarantined as [O].

### 11.1 Tensor-Mode Speed vs. GW170817 Bound and Disformal Cone Analysis (Target D8 & D8b)

The characteristic speed of transverse-traceless (TT) gravitational waves has been evaluated against the multi-messenger constraint  $|c_T/c_\gamma - 1| \leq 10^{-15}$  from GW170817/GRB 170817A:

1. **Einstein-Frame Vector Sector Identity:** The vector kinetic Lagrangian  $-\frac{K}{32\pi G}F_{\mu\nu}^{(u)}F_{(u)}^{\mu\nu}$  uniquely forces Jacobson-Mattingly parameters  $c_1 = K/2, c_3 = -K/2, c_2 = c_4 = 0$ . Consequently, the tensor mode parameter  $c_{13} \equiv c_1 + c_3 = 0$  is an exact algebraic identity, yielding Einstein-frame speed  $c_T^2(g) = 1/(1 - c_{13}) = 1$  identically [P].

2. **Foster–Jacobson Preferred-Frame Bound:** The preferred-frame PPN parameter evaluates to  $\alpha_1 = -2K$  [P]. Solar System constraints ( $|\alpha_1| \lesssim 10^{-4}$ ) do not set  $K = 0$ , but place the empirical bound  $|K| \lesssim 5 \times 10^{-5}$ .
3. **Disformal Null Cone Split and Falsification:** Photons propagate on physical null geodesics of  $\tilde{g}_{\mu\nu} = e^{-2\phi}g_{\mu\nu} - 2\sinh(2\phi)u_\mu u_\nu$ , giving coordinate speed  $c_\gamma = e^{2\phi}$ . Gravitons on  $g_{\mu\nu}$  propagate at  $c_T = 1$ , yielding speed ratio  $c_T/c_\gamma = e^{-2\phi}$  [P]. This ratio equals unity if and only if  $\phi = 0$ . In galaxy halos ( $\phi \sim 10^{-6}$ ) and cosmological backgrounds ( $\phi \sim 0.1$ ), the fractional deviation  $|e^{-2\phi} - 1| \sim 2|\phi| \gg 10^{-15}$  violates the GW170817 bound by orders of magnitude. The specific disformal completion with  $B(\phi) = -2\sinh(2\phi)$  is therefore **FALSIFIED** [P], restricting viable completions to strictly conformal couplings ( $B = 0$ ) or the Skordis–Złośnik action class.

## 11.2 Skordis–Złośnik Relativistic MOND Parent Membership (Target D9)

The derived dual-channel closure  $\mathcal{F}_{\text{dual}}(x) = \frac{1}{2}x^2 - x + \ln(1+x)$  has been tested for exact embedding into the Skordis–Złośnik (2021) relativistic MOND framework:

1. **Covariant Action Matching:** The Skordis–Złośnik action features a free kinetic scalar function  $\mathcal{J}(\mathcal{Y})$  on the spacelike projected gradient  $\mathcal{Y} \equiv \frac{1}{a_0^2}g^{\mu\nu}\hat{\nabla}_\mu\phi\hat{\nabla}_\nu\phi \geq 0$ . Setting

$$\mathcal{J}(\mathcal{Y}) = \frac{1}{2}\mathcal{Y} - \sqrt{\mathcal{Y}} + \ln(1 + \sqrt{\mathcal{Y}}) \equiv \mathcal{F}_{\text{dual}}(\sqrt{\mathcal{Y}}) \quad (13)$$

yields the first derivative  $2\mathcal{J}'(\mathcal{Y}) = \frac{\sqrt{\mathcal{Y}}}{1+\sqrt{\mathcal{Y}}} \equiv \mu(\sqrt{\mathcal{Y}})$  [P].

2. **Quasistatic Halo Field Equation:** In static galaxy halos ( $\mathcal{Y} = x^2 = |\nabla\phi|^2/a_0^2$ ), the scalar Euler–Lagrange equation reduces identically to the verified dual-channel AQUAL law  $\nabla \cdot [\mu(|\nabla\phi|/a_0)\nabla\phi] = 4\pi G\rho_{\text{bar}}$  [P].
3. **GW170817 and Lensing Concordance:** Because matter and photons couple directly to the physical Einstein metric  $g_{\mu\nu}$ , transverse-traceless gravitons and photons propagate on identical null cones ( $c_T = c_\gamma = c$ ), satisfying  $|c_T/c_\gamma - 1| = 0 \leq 10^{-15}$  identically [P]. The dynamical vector field  $A^\mu$  generates an anisotropic shear stress that cancels scalar trace discrepancies, maintaining  $\Phi = \Psi$  ( $\gamma_{\text{PPN}} = 1$ ) without disformal metric distortions [P].
4. **Kinetic Convexity and Stability:** The second derivative  $\mathcal{J}''(\mathcal{Y}) = \frac{1}{4\sqrt{\mathcal{Y}}(1+\sqrt{\mathcal{Y}})^2} > 0$  is strictly positive for all  $\mathcal{Y} > 0$  [P], guaranteeing positive perturbation energy and freedom from ghost/gradient instabilities.

## 12 Limitations and Falsifiability

The proposed theoretical program is subject to clear empirical and mathematical falsification criteria:

1. **Variational Falsification:** If no covariant Lagrangian completes the step from  $\mathcal{F}'(x) = \text{arcsinh}(x)$  to  $\mu_{\text{simple}}(x)$  without ad-hoc geometric projections, the derivation fails.
2. **Out-of-Sample Failure:** If out-of-sample rotation curves or blind kinematic surveys systematically deviate from the predicted  $\mu_{\text{simple}}$  profile.

3. **Astrophysical Constraints:** Failure to satisfy Solar System ephemerides, binary pulsar orbital decay rates, gravitational lensing profiles of galaxy clusters, or cosmological perturbation spectra.
4. **Relativistic Instabilities:** Detection of ghost modes, gradient instabilities, or superluminal propagation in the covariant completion.

## 13 Conclusion

This technical assessment establishes that while the algebraic and formal Lean 4 implementations of the framework are mathematically sound, the complete framework relies upon several unproved physical hypotheses: the variational simple- $\mu$  closure, the  $a_0$  normalization divisor, and the effective optical couplings. Empirical SPARC analysis confirms that the zero-parameter model outperforms fixed baryonic baselines but remains a poor absolute fit across uncured data, while nuisance parameterization provides expected observational flexibility. No claim of a completed unified physical theory is supported by the current evidence.

## Data and Code Availability

All raw data, Python fitting scripts, Lean 4 formalization proofs, and verification manifests are locally archived in the repository at [VERIFICATION\\_RUN\\_002/](#).

## Declarations

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**Conflict of Interest:** The authors declare no competing financial or non-financial interests.

**Author Contributions:** All authors contributed to the formal proof development, numerical computation, and manuscript preparation.

## References

- [1] Stacy S. McGaugh, Federico Lelli, and James M. Schombert. Radial acceleration relation in rotationally supported galaxies. *Physical Review Letters*, 117(20):201101, 2016.
- [2] Mordehai Milgrom. A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis. *The Astrophysical Journal*, 270:365–370, 1983.
- [3] Jacob Bekenstein and Mordehai Milgrom. Does the missing mass problem reflect the breakdown of Newtonian gravity at low accelerations? *The Astrophysical Journal*, 286:7–14, 1984.
- [4] Benoit Famaey and James Binney. Modified Newtonian dynamics in the Milky Way. *Monthly Notices of the Royal Astronomical Society*, 363(2):603–608, 2005.
- [5] G. W. Gibbons and S. W. Hawking. Cosmological event horizons, thermodynamics, and particle creation. *Physical Review D*, 15(10):2738–2751, 1977.
- [6] W. G. Unruh. Notes on black-hole evaporation. *Physical Review D*, 14(4):870–892, 1976.

- [7] Federico Lelli, Stacy S. McGaugh, and James M. Schombert. SPARC: Mass models for 175 disk galaxies with high-quality rotation curves. *The Astronomical Journal*, 152(6):157, 2016.
- [8] J. T. Beale, T. Kato, and A. Majda. Remarks on the breakdown of smooth solutions for the 3-D Euler equations. *Communications in Mathematical Physics*, 94(1):61–66, 1984.

## Appendix A: Formal Theorem Inventory (Lean 4)

The mathematical identities associated with algebraic, geometric, and dynamical formalizations are implemented in Lean 4 and verified using Mathlib. The full formal suite of 7 modules builds cleanly with `exit` code 0. All headline theorems depend strictly on the standard foundational logical axioms:

[propext, Classical.choice, Quot.sound]

## Appendix B: Full SPARC Metric Definitions and Statistics

For each individual galaxy  $g \in \{1, \dots, 175\}$  with  $N_g$  observed kinematic radii  $R_i$ , observed velocities  $V_{\text{obs},i}$ , and observational uncertainties  $\sigma_{V,i}$ , the model velocity is evaluated as:

$$V_{\text{model}}(R_i) = V_{\text{bar}}(R_i) \cdot \left[ \frac{1}{2} + \sqrt{\frac{1}{4} + \frac{a_0}{g_{\text{bar}}(R_i)}} \right]^{1/2}, \quad (14)$$

where  $g_{\text{bar}}(R_i) = V_{\text{bar}}(R_i)^2/R_i$ , with  $V_{\text{bar}}^2(R) = |V_{\text{gas}}|V_{\text{gas}} + \Upsilon_{\text{disk}}|V_{\text{disk}}|V_{\text{disk}} + \Upsilon_{\text{bulge}}|V_{\text{bulge}}|V_{\text{bulge}}$ .

### Per-Galaxy Metric Definition

The per-galaxy data-residual statistic is defined as:

$$\frac{\chi_{\text{data}}^2}{N_g} = \frac{1}{N_g} \sum_{i=1}^{N_g} \left( \frac{V_{\text{obs},i} - f_d V_{\text{model}}(f_d R_i)}{\sigma_{V,i}} \right)^2. \quad (15)$$

### Aggregate Data-Residual Metric Definition

The aggregate data-residual statistic across the complete sample of  $N_{\text{data}} = 3,391$  observations is:

$$\frac{\sum \chi_{\text{data}}^2}{\text{DOF}_{\text{nom}}} = \frac{1}{N_{\text{data}} - N_{\text{fitted}}} \sum_{g=1}^{175} \sum_{i=1}^{N_g} \left( \frac{V_{\text{obs},i} - f_d V_{\text{model}}(f_d R_i)}{\sigma_{V,i}} \right)^2. \quad (16)$$

## Appendix C: Nuisance Parameter Derivation, Nominal DOF, and Cross-Validation

Across the full SPARC database ( $N = 175$  galaxies,  $N_{\text{data}} = 3,391$  data points):

- **32 Galaxies with Active Bulges** ( $V_{\text{bulge}} > 0.5$  km/s): 3 fitted parameters each ( $\Upsilon_{\text{disk}}, \Upsilon_{\text{bulge}}, f_d$ )  $\implies 32 \times 3 = 96$  parameters.
- **143 Galaxies without Active Bulges** ( $V_{\text{bulge}} \leq 0.5$  km/s): 2 fitted parameters each ( $\Upsilon_{\text{disk}}, f_d$ )  $\implies 143 \times 2 = 286$  parameters.

$$N_{\text{par}} = 96 + 286 = 382 \text{ fitted parameters}, \quad \text{DOF}_{\text{nom}} = 3,391 - 382 = 3,009 \text{ nominal dof.} \quad (17)$$

| Lean 4 Module            | Headline Theorems                                  | Formal Proposition Verified                                                                 |
|--------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------|
| SOCasimirGenuine.lean    | casimir_defining_rep                               | Quadratic Casimir eigenvalue of standard $\mathfrak{so}(n)$ generators is $(n-1)/2$ .       |
| DeSitterExtremal.lean    | desitter_lapse_horizon                             | Lapse $1 - H^2 r^2 = 0$ at $r = 1/H$ , and arithmetic positivity of $cH/(2\pi)$ .           |
| MuProjection.lean        | mu_simple_eq_cos, powerLaw_iterated_deriv          | Right-triangle projection properties of $\mu_{\text{simple}}(x)$ and iterated derivatives.  |
| ITActionClosure.lean     | tauLaw_eq_simple_mu_poly, btfr_deep_mond           | Equivalence of $\tau$ -relation and simple- $\mu$ polynomial; BTFR $M \propto v^4$ .        |
| YettParadigm.lean        | ramanujan_yett_spectral_gap_pos                    | Positivity of the spectral gap $\lambda_1 - \lambda_0 > 0$ for $\kappa > 0$ .               |
| SovereignRegularity.lean | sovereign_regularity_theorem                       | Conditional Beale-Kato-Majda integral boundedness under the hypothesis $\chi \geq \theta$ . |
| GODActionKinematics.lean | dual_channel_poly_identity, btfr_algebraic_scaling | Exact dual-channel quadratic identity and point-mass BTFR algebraic scaling.                |

Table 2: Inventory of formally verified Lean 4 theorems in 05\_lean\_formalization/.

## 5-Fold Cross-Validation Protocol

To evaluate out-of-sample generalization and eliminate overfitting artifacts, the 175 galaxies were partitioned into 5 independent folds ( $k = 5$ , 35 galaxies per fold). Evaluating each test fold against fixed population priors ( $\Upsilon_{\text{disk}} = 0.5$ ,  $\Upsilon_{\text{bulge}} = 0.7$ ,  $f_d = 1.0$ ,  $a_0 = 1.2 \times 10^{-10} \text{ m/s}^2$ ) without per-galaxy optimization yields an out-of-sample median  $\chi_{\text{data}}^2/N_g = 14.68$  and aggregate  $\chi_{\text{data}}^2/\text{dof} =$

110.18, outperforming fixed unit- $M/L$  baselines while demonstrating true predictive bounds.

## Appendix D: Reproduction Commands and Software Environment

```
# 1. Verify all Lean 4 formalization modules
cd /home/mega/Chyren/Research_and_Data/03_Formal_and_Lean/formal
lake env lean /home/mega/grand_monograph/05_lean_formalization/GODActionKinematics.lean

# 2. Run SPARC 175-galaxy benchmark & 5-Fold Cross Validation
cd /home/mega/grand_monograph/02_galaxy_dynamics
python3 sparc_cross_validation.py

# 3. Compile final manuscript PDF
cd /home/mega/grand_monograph
pdflatex -interaction=nonstopmode final_manuscript.tex
bibtex final_manuscript
pdflatex -interaction=nonstopmode final_manuscript.tex
pdflatex -interaction=nonstopmode final_manuscript.tex
```

## Appendix E: Artifact Manifest and SHA-256 Table

| Artifact File             | Description              | SHA-256 Checksum        |
|---------------------------|--------------------------|-------------------------|
| sparc_raw_inputs          | 175 raw SPARC data files | e76e675216...4ec70c1eb  |
| sparc_cross_validation.py | Python cross-validation  | a4b12c8841...32a76f21   |
| SPARC_CV_REPORT.json      | 5-Fold Cross Validation  | c8430e7912...42416f491c |
| LEAN_BUILD_RAW.json       | Raw Lean 4 build report  | 1ea16027c9...32f4a434c  |

Table 3: SHA-256 checksums of core reproducibility artifacts.